

Arrival choices

Previous lesson reminder:

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

1 What caused the 2011 tsunami?

Support: Use the reminder strip or talk it through with an adult.

My response

2 Which meaning fits tectonic?

Support: Choose: Related to movement of Earth's plates OR People or things in harm's way.

My response

3 Why record the snapshot time?

Support: Use the model evidence anchor D or read one relevant card together.

My response

4 Can this small sample predict the next earthquake?

Support: Give a reason when ready.

My response

Stretch: use a precise example or explain which detail supports your answer.

Evidence cards

Read, point or listen. Keep the source label with any detail you use.

A Japan 11 March 2011

A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Source: USGS event record, official20110311054624120_30

Limit: Magnitude measures the earthquake; it is not a direct count of people affected.

B Effects and responses

Shaking and coastal flooding damaged places and disrupted people's lives. Earthquake-resistant design addresses shaking; warnings, evacuation routes and higher ground address tsunami exposure.

Source: Authored explanation of the case and risk-management principles

Limit: A sea wall or warning cannot eliminate all risk. Do not judge people without considering access and warning time.

C Heat and unequal risk

On 19 July 2022 the Met Office recorded 40.3°C at Coningsby, 40.2°C at St James's Park and 39.6°C at Topcliffe. Outdoor work increases exposure; difficulty accessing cool spaces can increase vulnerability. Shade and accessible cool spaces can reduce harm.

Source: Met Office July 2022 report, station table; risk explanation authored for teaching

Limit: This explains risk factors, not a medical assessment of an individual.

D Reading a hazard feed

A USGS event row can include UTC time, place, magnitude and depth in kilometres. A short feed is a dated snapshot. It cannot predict the next earthquake.

Source: USGS earthquake feed documentation

Limit: Small samples, reporting thresholds and later revisions affect patterns.

Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

Data for the enquiry

Archived USGS feed snapshot

UTC event time	Place	Magnitude	Depth km	Latitude	Longitude
2026-09-19T12:19:55+00:00	273 km NNW of Dêqên, China	4.5	10	32.0711	89.2243
2026-09-19T12:07:12+00:00	25 km NW of Ushibukamachi, Japan	4.6	10	32.3799	129.8568
2026-09-19T10:52:05+00:00	10 km ENE of Milpitas, CA	2.5	2.67	37.466667175293	-121.801834106445
2026-09-19T10:31:17+00:00	18 km SW of Mongar, Bhutan	4.8	10	27.1636	91.1016
2026-09-19T09:59:59+00:00	South Sandwich Islands region	5	77.62	-55.6255	-28.1116
2026-09-19T09:56:02+00:00	4 km NE of Límni, Greece	4.4	10	38.7884	23.3566

Units: Magnitude

Source: USGS M2.5+ past-day feed, generated 2026-09-19T14:08:13+00:00

Limit: This saved snapshot is not live at teaching time. Refresh explicitly or label the analysis as archived. A small selected sample cannot predict earthquakes.

My labelled graph or data notes

Shared investigation

1. Open the teacher-selected USGS feed and record its snapshot time.
2. Read magnitude and depth as different variables.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

Use the evidence cards and any visual sheet. Follow the two shared steps above, then record one decision.

Our explanation

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining.

Further challenge: Explain how threshold, sample duration or revisions can change the apparent pattern.

Supported independent task

Complete this route only. Track live hazards online and interpret data.

1. Capture a dated and timed live snapshot or use the explicitly dated archive if offline.
2. Enter a small event set in the table and graph one variable.
3. Identify a pattern and a conclusion limited to that snapshot.

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining. Complete the organiser one space at a time. Snapshot UTC _____. Source _____. Variable and units _____. Pattern _____. This cannot tell us _____.

Data source and date or capture time

Variables units and graph labels

A pattern supported by values

What the data cannot establish

An adult may read or scribe your exact response. They should not choose the answer.

Standard independent task

Complete this route only. Track live hazards online and interpret data.

1. Capture a dated and timed live snapshot or use the explicitly dated archive if offline.
2. Enter a small event set in the table and graph one variable.
3. Identify a pattern and a conclusion limited to that snapshot.

Support: Snapshot UTC _____. Source _____. Variable and units _____. Pattern _____. This cannot tell us _____.

Data source and date or capture time

Variables units and graph labels

A pattern supported by values

What the data cannot establish

Check the required evidence and revise one detail.

Stretch independent task

Complete this route only. Track live hazards online and interpret data.

1. Capture a dated and timed live snapshot or use the explicitly dated archive if offline.
2. Enter a small event set in the table and graph one variable.
3. Identify a pattern and a conclusion limited to that snapshot.
4. Explain how threshold, sample duration or revisions can change the apparent pattern.

Support: Choose the evidence independently. Use the frame only if it helps.

Data source and date or capture time

Variables units and graph labels

A pattern supported by values

What the data cannot establish

Explain how threshold, sample duration or revisions can change the apparent pattern.

Exit ticket

Choose a response mode. You may give your answers privately to an adult.

Why record the snapshot time?

My response

Can this small sample predict the next earthquake?

My response

Support: Snapshot UTC _____. Source _____. Variable and units _____. Pattern _____. This cannot tell us _____.

Further challenge: Explain how threshold, sample duration or revisions can change the apparent pattern.

One thing I changed or want to check
